

GROW computing Weeks 3 to 6

Four lessons of 40 minutes · Scratch maze continuation

These lessons follow the foundation work in Weeks 1 and 2. Pupils create a maze, build four controls, add a wall rule and make a win response. Use a pupil's own saved project as the starting point each week.

Week	Shared goal	AQA focus
3	Build the world	Outcome 1
4	Make controls reliable	Outcome 2
5	Make a rule work	Outcome 3
6	Finish and restart	Outcome 4

The lesson chassis

Stage	Minutes	Purpose
Arrival	0-3	Settle and locate the current file.
Starter	3-6	Retrieve and predict.
I Do	6-11	Watch a precise model.
We Do	11-18	Investigate and test together.
I Do 2	18-21	Model the next small part.
We Do 2 · Task	21-25	Rehearse the first independent action.
Independent Task	25-35	Create, test and explain an individual result.
Exit	35-40	Save, reopen and agree the next step.

Resources and access

Every week includes HTML, PowerPoint, a Main task booklet, Step by step booklet, Challenge booklet, teacher guide, PDFs and Scratch files. Choose and change routes from observed need. Quiet participation and spoken or pointed answers work in every route.

Evidence and time

Pupil creation is observed individually. Recovery files support access but cannot establish authorship of inherited content. The UAS lead records outcomes on AQA summary sheets. Week 6 reviews outcomes 1-4; Week 7 gameplay complexity and Week 8 full testing remain. Agree additional construction time at the readiness review when necessary.

[Current AQA unit 71638 Programming with Scratch unit 6 Level One](#)

Week 3 Build the world

One 40-minute lesson

I can create a maze, a player and a finish as three sprites.

Words: sprite, backdrop, route, finish

Stage and time	Teaching and pupil action
Arrival 0-3	Open W03_Blank.sb3. Recall your mission choice. Today you will create three sprites.
Starter 3-6	Look at the small player and the large player. Which can fit through the gap? Point to what you would change.
I Do 6-11	Maze holds the walls. Player moves around the route. Finish marks the goal.
We Do 11-18	Trace a route together. Try a larger player. Predict the problem. Choose one repair and test it.
I Do 2 18-21	Place Player away from a wall. Place Finish where Player can reach it. Look at the start x and y.
We Do 2 · Task 21-25	Name each of the three sprites. Check the first passage with a token. Agree one useful layout change.
Independent Task 25-35	Create Maze, Player and Finish. Use your task route for the next action. Check that Player fits through your route.
Exit 35-40	Save as W03_Maze.sb3. Load it and select each sprite. Show what your design idea changed.

Step by step: Create one sprite at a time using the role cards.

Main task: Create three sprites and check a clear route.

Challenge: Compare player sizes and repair a narrow gap.

Voice: Which passage, object or mission detail should change? Choose one useful layout change.

Evidence: Observe each pupil creating and selecting all three sprites. Record what they did and the support used. A downloaded finished maze does not establish creation.

Prepare: Start a new project with W03_Blank.sb3. Have the named booklet, a teacher demo and the approved save location ready.

Week 4 Make controls reliable

One 40-minute lesson

I can create and test all four arrow controls in my maze.

Words: event, x, y, reset

Stage and time	Teaching and pupil action
Arrival 0-3	Open your own W03_Maze.sb3. Select Player. Point to its safe starting place.
Starter 3-6	The up key changes x by 10. Predict the direction Player moves. Which block would you inspect?
I Do 6-11	Right and left use x. Use a separate event for each key. Set a clear green-flag start.
We Do 11-18	Choose the correct axis for up. Predict, then test the repair. Reset before each new test.
I Do 2 18-21	Up and down use y. Up adds 10. Down adds -10. Use change y in each script.
We Do 2 · Task 21-25	Reset Player. Predict right, up, left, down. Test and check the final position.
Independent Task 25-35	Build all four key scripts on Player. Test each direction from reset. Explain one block and its effect.
Exit 35-40	Save as W04_Controls.sb3. Load it and test all four directions. Show which support or idea helped.

Step by step: Build and test one key at a time.

Main task: Build four keys and test a return to the start.

Challenge: Test whether sprite direction changes x/y controls.

Voice: Which key or action needs clearer help? Choose a useful support or step-size change and test it.

Evidence: Observe pupil creation of all four arrow scripts and a practical demonstration. Check any previously inherited controls by having the pupil construct the relevant script.

Prepare: Open your own W03_Maze.sb3. Have the named booklet, a teacher demo and the approved save location ready.

Week 5 Make a rule work

One 40-minute lesson

I can make Player return to the start when it touches Maze.

Words: condition, touching, consequence, loop

Stage and time	Teaching and pupil action
Arrival 0-3	Open your own W04_Controls.sb3. Check one arrow key. Find a start clear of Maze.
Starter 3-6	The script checks only at the green flag. Player touches a wall later. Will the script check again?
I Do 6-11	Keep checking while the game runs. Ask: touching Maze? If true, go to the safe start.
We Do 11-18	Predict the result in clear space. Test one wall, then a different wall. Compare one check with repeated checks.
I Do 2 18-21	Put if then inside forever. Put touching Maze? in its question space. Put the return action inside if then.
We Do 2 · Task 21-25	Point to the condition. Point to the action that follows contact. Check the chosen start is clear.
Independent Task 25-35	Build the wall rule on Player. Test two walls and one clear space. Make one repair if a result is wrong.
Exit 35-40	Save as W05_Walls.sb3. Load it and test a wall again. Show a pupil idea and its effect.

Step by step: Use the nesting guide and build one block at a time.

Main task: Build the repeated wall rule and test two walls.

Challenge: Investigate and repair a start inside a wall.

Voice: Is the return point clear and comfortable to use? Suggest a fair response or a clearer way to show it.

Evidence: Observe the pupil constructing the collision script, then demonstrate contact at two wall positions. Record the actual consequence and any prompts.

Prepare: Open your own W04_Controls.sb3. Have the named booklet, a teacher demo and the approved save location ready.

Week 6 Finish and restart

One 40-minute lesson

I can create a win response and start a fresh game.

Words: win condition, message, wait until, restart

Stage and time	Teaching and pupil action
Arrival 0-3	Open your own W05_Walls.sb3. Test the wall rule. Point to Finish.
Starter 3-6	Player reaches Finish. What should the game do? Choose a clear, quiet win signal.
I Do 6-11	Wait until Player touches Finish. Show a short win message. This win script then reaches its end.
We Do 11-18	Predict away from Finish and at Finish. Test the win, then check again. Compare one response with repeated messages.
I Do 2 18-21	First clear the old message. Go to a safe start before waiting for Finish. The green flag begins a new run.
We Do 2 · Task 21-25	Win once. Click the green flag. Check the message clears. Test the wall rule and win again.
Independent Task 25-35	Create the win script on Player. Test a win and a fresh start. Keep the controls and wall rule working.
Exit 35-40	Save as W06_Win.sb3. Load it and show a win or restart test. Agree the next step with your teacher.

Step by step: Use the win reference and build one part at a time.

Main task: Build a win response and test a fresh run.

Challenge: Repair repeated win behaviour and retest the game.

Voice: Choose a short win message or a quiet visual signal. Test whether it is clear and comfortable.

Evidence: Observe pupil construction and demonstration of the win script. Record a win and a restarted run. Audit outcomes 1-4 individually; do not claim outcomes 5-6 from the teacher model.

Prepare: Open your own W05_Walls.sb3. Have the named booklet, a teacher demo and the approved save location ready.